Statistical Foundations of Learning

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Winter Semester 2026/27

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1 Statistical learning problem and Bayes risk

1.1 Learning problem and risk

Setup. Inputs live in a **feature space** $\mathcal{X}$, outputs in a **label set** $\mathcal{Y}$ (binary classification: $\mathcal{Y} = \{0, 1\}$ or $\{\pm 1\}$; regression: $\mathcal{Y} \subseteq \mathbb{R}$). A **predictor** is a map $h : \mathcal{X} \to \mathcal{Y}$. Given a **training sample** $S = ((x_i, y_i))_{i=1}^m \in (\mathcal{X} \times \mathcal{Y})^m$, a **learner** $\mathcal{A}$ returns a predictor $\hat{h}_S = \mathcal{A}(S)$; it must accept samples of every size $m \geq 1$ (think of a neural network trained by gradient descent on S).

Data model. Learning only makes sense if test data resemble training data. We assume a fixed but unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$: training and test pairs are all drawn from $\mathcal{D}$, independently, so $S \sim \mathcal{D}^m$. Equivalently, $\mathcal{D}$ is specified by the marginal $\mathcal{D}_\mathcal{X}$ of x and the conditional law $\mathbb{P}(y \mid x)$ of the label. A deterministic labelling $y = f(x)$ is the special case of a degenerate conditional; it cannot express that two equal inputs may carry different labels.

Loss and risk. A **loss** $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$ penalises a prediction $h(x)$ when the truth is y. The two standard choices are the **0-1 loss** $\mathbb{1}\{h(x) \neq y\}$ for classification and the **squared loss** $(h(x) - y)^2$ for regression.

Definition 1.1 (Risk)**.** The **risk** (generalisation error) of h is its average loss on a new pair from $\mathcal{D}$,

$$L_\mathcal{D}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h(x), y)\big].$$

For a learner, the quantity of interest is $L_\mathcal{D}(\hat{h}_S)$, the expected test error of the predictor learned from S.

Under the 0-1 loss, $L_\mathcal{D}(h)$ is the probability that h misclassifies, $L_\mathcal{D}(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(h(x) \neq y)$.

Example 1.2 (Two Gaussian classes)**.** Take $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{\pm 1\}$, both labels with probability $\frac{1}{2}$, and class-conditional laws $\mathcal{N}(-\mu, \sigma^2)$ for $y = -1$ and $\mathcal{N}(\mu, \sigma^2)$ for $y = +1$, with $\mu > 0$. Then $\mathcal{D}_\mathcal{X} = \frac{1}{2}\mathcal{N}(-\mu, \sigma^2) + \frac{1}{2}\mathcal{N}(\mu, \sigma^2)$, and $\mathbb{P}(y = +1 \mid x)$ increases from 0 to 1, crossing $\frac{1}{2}$ at $x = 0$. A threshold rule $h(x) = \operatorname{sign}(x - t)$ errs on the $+1$ points left of t and on the -1 points right of t (fig. 1); moving t away from 0 trades errors on one class for errors on the other. The optimal threshold is $t = 0$, where $\operatorname{sign}(x)$ always predicts the more likely label (theorem 1.6); in practice it has to be found from S alone.

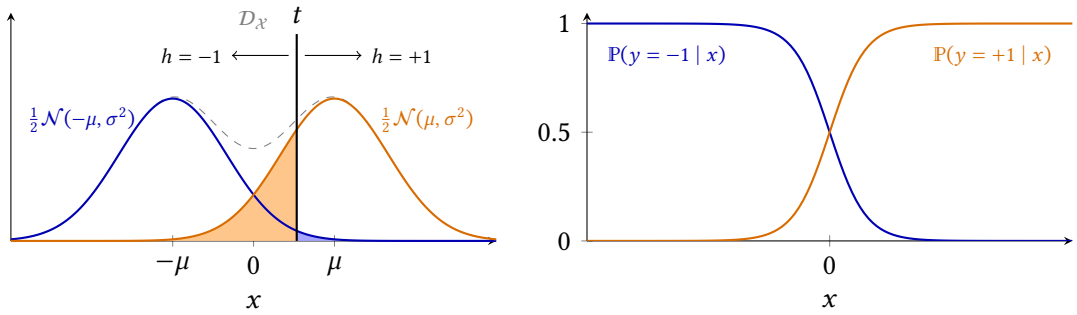


Figure 1: Left: weighted class densities, $\mathcal{D}_\mathcal{X}$ (dashed), errors of $\operatorname{sign}(x - t)$ (shaded). Right: $\mathbb{P}(y = \pm 1 \mid x)$.

1.2 Empirical risk minimisation

Ideally we would solve $\min_{h \in \mathcal{Y}^\mathcal{X}} L_\mathcal{D}(h)$, but $L_\mathcal{D}(h)$ cannot be evaluated without knowing $\mathcal{D}$, so it is replaced by its sample counterpart.

Definition 1.3 (Empirical risk, ERM)**.** The **empirical risk** (training error) of h on S is $L_S(h) = \frac{1}{m}\sum_{i=1}^m \ell(h(x_i), y_i)$. **Empirical risk minimisation** (ERM) solves $\min_{h \in \mathcal{Y}^\mathcal{X}} L_S(h)$.

For fixed h, $L_S(h)$ is an unbiased estimate of $L_D(h)$: each $(x_i, y_i) \sim D$, so by linearity $\mathbb{E}_{S\sim D^m}[L_S(h)] = \frac{1}{m}\sum_{i=1}^m \mathbb{E}\big[\ell(h(x_i), y_i)\big] = L_D(h)$. Minimising it over *all* functions, however, can overfit badly:

Example 1.4 (ERM can fail)**.** Let $x \sim \text{Uniform}([0,1])$ and $y = \mathbb{1}\{x > 0.1\}$. The predictor that outputs 1 on the training points labelled 1 and 0 everywhere else has $L_S(\hat{h}) = 0$. A new x falls in the finite set $\{x_1, \dots, x_m\}$ with probability 0, so $\hat{h}(x) = 0$ almost surely and $\hat{h}$ errs exactly when $y = 1$: $L_D(\hat{h}) = \mathbb{P}(y = 1) = \mathbb{P}(x > 0.1) = 0.9$.

1.3 Bayes risk

Suppose now that D is known. The **Bayes risk** is $L_D^* = \min_{h\in\mathcal{Y}^{\mathcal{X}}} L_D(h)$, and a minimiser h^* is a **Bayes predictor**.

Classification. Let $\mathcal{Y} = \{0,1\}$, ℓ the 0-1 loss, and write $\eta(x) = \mathbb{P}(y = 1 \mid x)$, so that $\mathbb{P}(y = 0 \mid x) = 1 - \eta(x)$.

Theorem 1.5 (Risk of a binary classifier)**.** *For every deterministic $h : \mathcal{X} \to \{0,1\}$,*

$$L_D(h) = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\mathbb{P}(h(x) \neq y \mid x)\big] = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}\big].$$

Proof. By the tower property,

$$L_D(h) = \mathbb{E}_{(x,y)}\big[\mathbb{1}\{h(x) \neq y\}\big] = \mathbb{E}_x\Big[\mathbb{E}_{y|x}\big[\mathbb{1}\{h(x) \neq y\}\big]\Big] = \mathbb{E}_x\big[\mathbb{P}(h(x) \neq y \mid x)\big].$$

An error occurs either when $h(x) = 0,\ y = 1$ or when $h(x) = 1,\ y = 0$, so $\mathbb{1}\{h(x) \neq y\} = \mathbb{1}\{h(x) = 0\}\mathbb{1}\{y = 1\} + \mathbb{1}\{h(x) = 1\}\mathbb{1}\{y = 0\}$. Given x, $h(x)$ is a constant (h is deterministic), so only the indicators of y are averaged, with $\mathbb{E}_{y|x}[\mathbb{1}\{y = 1\}] = \eta(x)$ and $\mathbb{E}_{y|x}[\mathbb{1}\{y = 0\}] = 1 - \eta(x)$:

$$\mathbb{P}(h(x) \neq y \mid x) = \eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}. \qquad \square$$

At each x, a classifier therefore pays $\eta(x)$ by predicting 0 or $1 - \eta(x)$ by predicting 1. The best choice is the cheaper option, which defines the **Bayes classifier**

$$h^*(x) = \mathbb{1}\{\eta(x) \geq \tfrac{1}{2}\}.$$

Theorem 1.6 (Bayes risk equals the risk of the Bayes classifier)**.** *For the 0-1 loss,*

$$L_D^* = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\min\{\eta(x), 1 - \eta(x)\}\big] = L_D(h^*) = \min_{h\in\{0,1\}^{\mathcal{X}}} L_D(h).$$

Proof. Fix x. Exactly one of $\mathbb{1}\{h(x) = 0\}$, $\mathbb{1}\{h(x) = 1\}$ equals 1, so the integrand of theorem 1.5 is either $\eta(x)$ or $1-\eta(x)$, hence $\geq \min\{\eta(x), 1-\eta(x)\}$. Taking $\mathbb{E}_x$, every h satisfies $L_D(h) \geq \mathbb{E}_x[\min\{\eta(x), 1-\eta(x)\}]$. For h^*: if $\eta(x) \geq \frac{1}{2}$ then $h^*(x) = 1$ and the integrand is $1 - \eta(x) = \min\{\eta(x), 1 - \eta(x)\}$; if $\eta(x) < \frac{1}{2}$ then $h^*(x) = 0$ and it is $\eta(x) = \min\{\eta(x), 1 - \eta(x)\}$. So h^* attains the lower bound, which is therefore the minimum. $\square$

Example 1.7 (Installation crash)**.** Let $x \sim \text{Uniform}[0,1]$ be the completed fraction of a software installation, with $y = 1$ meaning that the system crashes after the fraction x has been completed, and $\eta(x) = |1 - 2x|$. Then $\eta(x) \geq \frac{1}{2}$ iff $x \leq \frac{1}{4}$ or $x \geq \frac{3}{4}$, so $h^* = 1$ on $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$ and $h^* = 0$ on $(\frac{1}{4}, \frac{3}{4})$. The function $\min\{\eta, 1 - \eta\}$ is symmetric about $x = \frac{1}{2}$, and on $[0, \frac{1}{2}]$ (where $\eta = 1 - 2x$, $1 - \eta = 2x$) it equals $2x$ for $x \leq \frac{1}{4}$ and $1 - 2x$ for $x \geq \frac{1}{4}$. Hence

$$L_D^* = L_D(h^*) = 2\Big(\int_0^{1/4} 2x\,dx + \int_{1/4}^{1/2} (1 - 2x)\,dx\Big) = 2\Big(\frac{1}{16} + \frac{1}{16}\Big) = \frac{1}{4}.$$

Regression. Let $\mathcal{Y} = \mathbb{R}$ with the squared loss, so that $L_D(h) = \mathbb{E}[(h(x) - y)^2]$. ERM is then least-squares regression (ordinary least squares when h is restricted to linear functions).

Proposition 1.8 (Bayes regressor)**.** *For the squared loss, $h^*(x) = \mathbb{E}[y \mid x]$ and $L_D^* = \mathbb{E}_x\big[\operatorname{Var}(y \mid x)\big]$.*

Proof. Fix x; for deterministic h, $h(x)$ is a constant given x. Expanding the square and then completing it,

$$\begin{aligned}\mathbb{E}\big[(h(x) - y)^2 \mid x\big] &= h(x)^2 - 2h(x)\,\mathbb{E}[y \mid x] + \mathbb{E}[y^2 \mid x] \\ &= \big(h(x) - \mathbb{E}[y \mid x]\big)^2 + \underbrace{\mathbb{E}[y^2 \mid x] - (\mathbb{E}[y \mid x])^2}_{=\operatorname{Var}(y|x)}.\end{aligned}$$

Taking $\mathbb{E}_x$ gives $L_D(h) = \mathbb{E}_x\big[(h(x) - \mathbb{E}[y \mid x])^2\big] + \mathbb{E}_x[\operatorname{Var}(y \mid x)]$. The second term does not depend on h; the first is ≥ 0 and equals 0 for $h(x) = \mathbb{E}[y \mid x]$. □

Example 1.9 (Signal plus noise)**.** If $y = f(x) + \epsilon$ with ϵ independent of x, $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon^2] = \sigma^2$, then

$$\mathbb{E}[y \mid x] = f(x) + \mathbb{E}[\epsilon] = f(x), \qquad \operatorname{Var}(y \mid x) = \operatorname{Var}(\epsilon) = \mathbb{E}[\epsilon^2] - (\mathbb{E}[\epsilon])^2 = \sigma^2,$$

so by proposition 1.8 $h^* = f$ and $L_D^* = \sigma^2$.

2 Risk of linear regression

2.1 Least squares and the linear model

Ordinary least squares. Let $\mathcal{X} = \mathbb{R}^p$, $\mathcal{Y} = \mathbb{R}$ and $S = ((x_i, y_i))_{i=1}^m \subset \mathbb{R}^p \times \mathbb{R}$. A **linear model** predicts $h_w(x) = x^\top w$ with $w \in \mathbb{R}^p$. The **ordinary least squares** (OLS) estimator is ERM with the squared loss over linear models,

$$\hat{w} = \operatorname*{arg\,min}_{w \in \mathbb{R}^p} L_S(h_w), \qquad L_S(h_w) = \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2.$$

Writing $\hat{\Sigma} = \frac{1}{m}\sum_{i=1}^m x_i x_i^\top$ (a symmetric positive semidefinite $p \times p$ matrix), the minimiser is, whenever $\hat{\Sigma}$ is invertible,

$$\hat{w} = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big).$$

Ridge regression. When $\hat{\Sigma}$ is singular, the **ridge** estimator with parameter $\lambda > 0$ replaces it by $\hat{\Sigma} + \lambda I$, whose eigenvalues are all $\geq \lambda$, so that it is always invertible. Equivalently, ridge is ERM penalised by the squared Euclidean norm of w:

$$\hat{w}_\lambda = \big(\hat{\Sigma} + \lambda I\big)^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big) = \operatorname*{arg\,min}_{w \in \mathbb{R}^p}\ \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2 + \lambda\|w\|^2.$$

Data model. Since y is continuous, its conditional law given x is specified through a model rather than through the probabilities of single values. Let $\mathcal{D}_\mathcal{X}$ be a distribution on $\mathbb{R}^p$ with $\Sigma = \mathbb{E}_x[xx^\top]$, fix $w^* \in \mathbb{R}^p$, and set

$$y = x^\top w^* + \epsilon, \qquad \epsilon \text{ independent of } x, \quad \mathbb{E}[\epsilon] = 0, \quad \mathbb{E}[\epsilon^2] = \sigma^2.$$

Then $\mathbb{E}[y \mid x] = x^\top w^*$ and $\operatorname{Var}(y \mid x) = \sigma^2$, so by proposition 1.8 the Bayes predictor is h_{w^*} and $L_D^* = \sigma^2$. For $S \sim \mathcal{D}^m$ we write $y_i = x_i^\top w^* + \epsilon_i$, where $\epsilon_1, \dots, \epsilon_m$ are i.i.d. copies of ϵ, independent of $x_1, \dots, x_m$. The estimator $\hat{w}$ is a function of S, hence random, and the quantity to compute is the expected risk

$$\mathbb{E}_S\big[L_D(h_{\hat{w}})\big] = \mathbb{E}_S\Big[\mathbb{E}_{(x,y)\sim\mathcal{D}}\big[(y - x^\top \hat{w})^2\big]\Big],$$

with the test pair $(x, y) \sim \mathcal{D}$ independent of S.

2.2 Bias-variance decomposition

Notation. For a positive semidefinite Σ let $\|a\|_\Sigma^2 = a^\top \Sigma a$, and let $\mathrm{trace}(M) = \sum_i M_{ii}$. The covariance matrix of the estimator is

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - \mathbb{E}_S[\hat{w}])(\hat{w} - \mathbb{E}_S[\hat{w}])^\top\right].$$

Theorem 2.1 (Risk of a learned linear model)**.** *Under the data model above, let $\hat{w}$ be learned from $S \sim \mathcal{D}^m$ by any learner. Then*

$$\begin{aligned}\mathbb{E}_S\left[L_{\mathcal{D}}(h_{\hat{w}})\right] &= \mathbb{E}_S\left[\|\hat{w} - w^*\|_\Sigma^2\right] + \sigma^2 \\ &= \underbrace{\|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2}_{\textit{bias}} + \underbrace{\mathbb{E}_S\left[\|\hat{w} - \mathbb{E}_S[\hat{w}]\|_\Sigma^2\right]}_{\textit{variance}} + \underbrace{\sigma^2}_{\textit{Bayes risk}} \\ &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2.\end{aligned}$$

Proof. First equality. Fix S, so that $\hat{w}$ is a constant for the independent test pair. Inserting the model and expanding the square, with $(x^\top a)^2 = a^\top x x^\top a$,

$$\begin{aligned}L_{\mathcal{D}}(h_{\hat{w}}) &= \mathbb{E}_{x,\epsilon}\left[(x^\top \hat{w} - x^\top w^* - \epsilon)^2\right] = \mathbb{E}_{x,\epsilon}\left[(x^\top(\hat{w} - w^*) - \epsilon)^2\right] \\ &= \mathbb{E}_x\left[(\hat{w} - w^*)^\top x x^\top (\hat{w} - w^*)\right] - 2\,\mathbb{E}_{x,\epsilon}\left[\epsilon\, x^\top(\hat{w} - w^*)\right] + \mathbb{E}[\epsilon^2] && \text{(linearity)} \\ &= (\hat{w} - w^*)^\top \mathbb{E}_x[x x^\top](\hat{w} - w^*) - 2\,\mathbb{E}[\epsilon]\,\mathbb{E}_x\left[x^\top(\hat{w} - w^*)\right] + \sigma^2 && (x,\ \epsilon \text{ independent}) \\ &= \|\hat{w} - w^*\|_\Sigma^2 + \sigma^2 && (\mathbb{E}[\epsilon] = 0).\end{aligned}$$

Taking $\mathbb{E}_S$ on both sides gives the claim.

Second equality. Split $\hat{w} - w^* = u + v$ with $u = \hat{w} - \mathbb{E}_S[\hat{w}]$ random and $v = \mathbb{E}_S[\hat{w}] - w^*$ deterministic. Since Σ is symmetric,

$$\|u + v\|_\Sigma^2 = u^\top \Sigma u + v^\top \Sigma v + 2\, u^\top \Sigma v.$$

By linearity $\mathbb{E}_S[u^\top \Sigma v] = \mathbb{E}_S[u]^\top \Sigma v = 0$, because $\mathbb{E}_S[u] = \mathbb{E}_S[\hat{w}] - \mathbb{E}_S[\hat{w}] = 0$. Hence $\mathbb{E}_S\|\hat{w} - w^*\|_\Sigma^2 = \mathbb{E}_S\|u\|_\Sigma^2 + \|v\|_\Sigma^2$, which is variance plus bias.

Third equality. A scalar equals its trace, and the trace is cyclic and commutes with expectation (both are linear):

$$\mathbb{E}_S\left[u^\top \Sigma u\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u^\top \Sigma u)\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u u^\top \Sigma)\right] = \mathrm{trace}\left(\mathbb{E}_S[u u^\top]\,\Sigma\right) = \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right). \quad \square$$

Bias and variance are ≥ 0 (Σ is positive semidefinite), so no learned linear model beats the Bayes risk σ^2, which h_{w^*} attains. An **unbiased** estimator, $\mathbb{E}_S[\hat{w}] = w^*$, removes the bias term; still, a biased estimator with small enough $\mathrm{Cov}_S(\hat{w})$ can have a smaller expected risk. Balancing the two terms is the **bias-variance tradeoff**.

2.3 Risk of OLS and consistency

From now on $\hat{w}$ is the OLS estimator.

Theorem 2.2 (OLS is unbiased)**.** *If $\hat{\Sigma}$ is invertible with probability* 1*, then $\mathbb{E}_S[\hat{w}] = w^*$.*

Proof. Given $x_1, \ldots, x_m$, the matrix $\hat{\Sigma}$ is fixed, and $\mathbb{E}[y_i \mid x_1, \ldots, x_m] = x_i^\top w^* + \mathbb{E}[\epsilon_i] = x_i^\top w^*$ because ϵ_i is independent of the inputs and has mean 0. Therefore

$$\begin{aligned}\mathbb{E}_S[\hat{w}] &= \mathbb{E}_{x_1,\ldots,x_m}\left[\mathbb{E}\left[\hat{w} \mid x_1, \ldots, x_m\right]\right] && \text{(tower property)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i\,\mathbb{E}[y_i \mid x_1, \ldots, x_m]\right)\right] && \text{(linearity)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\right) w^*\right] = \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\hat{\Sigma}\, w^*\right] = w^*. && \square\end{aligned}$$

Theorem 2.3 (Covariance of the OLS estimator). *If $\hat{\Sigma}$ is invertible with probability 1, then*

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].$$

Proof. The first equality is the definition of Cov_S with $\mathbb{E}_S[\hat{w}] = w^*$ (theorem 2.2). Substituting $y_i = x_i^\top w^* + \epsilon_i$,

$$\frac{1}{m}\sum_{i=1}^m y_i x_i = \Big(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\Big) w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i = \hat{\Sigma} w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i \quad\Longrightarrow\quad \hat{w} - w^* = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m x_i \epsilon_i\Big).$$

Since $\hat{\Sigma}^{-1}$ is symmetric, $(\hat{w} - w^*)^\top = \big(\frac{1}{m}\sum_j x_j^\top \epsilon_j\big)\hat{\Sigma}^{-1}$, so

$$(\hat{w} - w^*)(\hat{w} - w^*)^\top = \frac{1}{m^2}\,\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \epsilon_i \epsilon_j\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}.$$

The ϵ_i are i.i.d. and independent of the inputs, so $\mathbb{E}[\epsilon_i \epsilon_j \mid x_1, \dots, x_m] = \mathbb{E}[\epsilon_i \epsilon_j]$, which is $\mathbb{E}[\epsilon_i]\mathbb{E}[\epsilon_j] = 0$ for $i \neq j$ and $\mathbb{E}[\epsilon_i^2] = \sigma^2$ for $i = j$. Conditioning on the inputs (tower property) and using linearity,

$$\begin{aligned}
\mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] &= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \mathbb{E}[\epsilon_i\epsilon_j]\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}\Big] \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sigma^2 \sum_{i=1}^m x_i x_i^\top\Big)\hat{\Sigma}^{-1}\Big] \qquad \text{(only } i = j \text{ survives)} \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\big(\sigma^2 m \hat{\Sigma}\big)\hat{\Sigma}^{-1}\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].
\end{aligned}$$

□

Gaussian features. Let $\mathcal{D}_\mathcal{X} = \mathcal{N}(0, \Sigma)$ with $\Sigma \in \mathbb{R}^{p\times p}$ invertible (the mean is 0, so $\Sigma = \mathbb{E}[xx^\top]$ as in the data model), and $y = x^\top w^* + \epsilon$ as before. We use without proof the following facts on the Wishart distribution, for $x_1, \dots, x_m$ i.i.d. $\mathcal{N}(0, \Sigma)$. If $m \geq p$, then $\hat{\Sigma}$ is invertible with probability 1; $\hat{\Sigma}$ is Wishart with m degrees of freedom and scale matrix $\frac{1}{m}\Sigma$; $\hat{\Sigma}^{-1}$ is inverse Wishart with m degrees of freedom and scale matrix $m\Sigma^{-1}$; and for $m > p + 1$,

$$\mathbb{E}\left[\hat{\Sigma}^{-1}\right] = \frac{m}{m - p - 1}\,\Sigma^{-1}.$$

Theorem 2.4 (Risk of OLS for Gaussian features and additive noise). *For $\mathcal{D}$ as above and $m > p + 1$,*

$$\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] = \underbrace{\frac{\sigma^2 p}{m - p - 1}}_{\text{variance}} + \underbrace{\sigma^2}_{\text{Bayes risk}} = \frac{\sigma^2(m-1)}{m - p - 1}.$$

In particular, for fixed p, $\mathbb{E}_S[L_\mathcal{D}(h_{\hat{w}})] \to \sigma^2 = L_\mathcal{D}^*$ as $m \to \infty$.

Proof. Since $m > p + 1 \geq p$, $\hat{\Sigma}$ is invertible with probability 1, so theorems 2.2 and 2.3 apply. Starting from theorem 2.1,

$$\begin{aligned}
\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2 \\
&= 0 + \mathrm{trace}\left(\frac{\sigma^2}{m}\,\mathbb{E}\left[\hat{\Sigma}^{-1}\right]\Sigma\right) + \sigma^2 && \text{(theorems 2.2 and 2.3)} \\
&= \mathrm{trace}\left(\frac{\sigma^2}{m}\cdot\frac{m}{m - p - 1}\,\Sigma^{-1}\Sigma\right) + \sigma^2 && \text{(Wishart mean)} \\
&= \frac{\sigma^2}{m - p - 1}\,\mathrm{trace}(I_p) + \sigma^2 = \frac{\sigma^2 p}{m - p - 1} + \sigma^2 && (\mathrm{trace}(I_p) = p) \\
&= \frac{\sigma^2(p + m - p - 1)}{m - p - 1} = \frac{\sigma^2(m-1)}{m - p - 1}.
\end{aligned}$$

For fixed p, $\frac{m-1}{m-p-1} \to 1$ as $m \to \infty$. □

Consistency. Let $\mathcal{D}$ be a distribution on $\mathcal{X} \times \mathcal{Y}$, ℓ a loss with Bayes risk $L_{\mathcal{D}}^*$, and $\hat{h}_S$ the predictor that a learner $\mathcal{A}$ returns on $S \sim \mathcal{D}^m$.

Definition 2.5 (Consistency)**.** $\mathcal{A}$ is **consistent** for $\mathcal{D}$ and ℓ if $\mathbb{E}_{S\sim\mathcal{D}^m}\left[L_{\mathcal{D}}(\hat{h}_S)\right] \to L_{\mathcal{D}}^*$ as $m \to \infty$. It is **universally consistent** if it is consistent for every distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$.

By theorem 2.4, OLS is consistent for $x \sim \mathcal{N}(0, \Sigma)$, $y = x^\top w^* + \epsilon$ (with p fixed). It is not universally consistent: it fails as soon as the true model is not linear, for instance $y = (x^\top w^*)^2 + \epsilon$.

Remark. Take $x \sim \mathcal{N}(0, \Sigma)$ and $w^* \neq 0$ in this example. Then $\mathbb{E}[y \mid x] = (x^\top w^*)^2$ and $\mathrm{Var}(y \mid x) = \sigma^2$, so $L_{\mathcal{D}}^* = \sigma^2$. For any w put $Z = x^\top w - (x^\top w^*)^2$; the decomposition in the proof of proposition 1.8, $\mathbb{E}[Z^2] \geq (\mathbb{E}Z)^2$, $\mathbb{E}[x] = 0$ and $\mathbb{E}[(x^\top w^*)^2] = w^{*\top}\mathbb{E}[xx^\top]w^*$ give

$$L_{\mathcal{D}}(h_w) = \mathbb{E}[Z^2] + \sigma^2 \geq (\mathbb{E}Z)^2 + \sigma^2 = \left(0 - \|w^*\|_\Sigma^2\right)^2 + \sigma^2 = \|w^*\|_\Sigma^4 + \sigma^2.$$

Hence $\mathbb{E}_S[L_{\mathcal{D}}(h_{\hat{w}})] \geq L_{\mathcal{D}}^* + \|w^*\|_\Sigma^4$ for every m, and $\|w^*\|_\Sigma > 0$ since Σ is invertible.